Waves and Optics

- Week 5: interference
- Two-slit experiment
- Coherence and path difference

Two-slit interference

- Bright fringes where the path difference is a whole number of wavelengths
- Fringe spacing grows with distance to the screen
- Shorter wavelength, closer fringes

Fringes on the screen

Summary

- Superposition explains interference
- Measure fringe spacing to find the wavelength
- Next week: diffraction